

CT5102 Assignment 3 - Data Frames and Plotting

October 22, 2025

Table of contents

Objective	1
Write a filter function to filter an atomic vector of characters (25 marks)	1
Use <code>lapply()</code> to generate summary stats for variables in <code>mtcars</code> (25 marks)	2
Process and visualise the data from <code>eirgrid17</code> (25 marks)	3
Process and visualise the data from <code>observations</code> (25 marks)	5

Objective

The aim of this assignment is to process data frames using the pipe operator, and also writing useful utility functions. Plotting of results is also an important goal.

Write a filter function to filter an atomic vector of characters (25 marks)

First, write a function `filter_names()` that matches the first letter of a character vector to a target character, and returns all values that start with this letter. Use the function `substr()` to extract the first element of a character vector.

For example, here is how it should work. Internally, the function should use `lapply()`.

```
v <- names(mtcars)
v
```

```
[1] "mpg"  "cyl"  "disp" "hp"    "drat" "wt"    "qsec" "vs"    "am"    "gear"
[11] "carb"
```

```
nv <- filter_names(v,"d")
nv
```

```
[1] "disp" "drat"
```

Use lapply() to generate summary stats for variables in mtcars (25 marks)

Write lapply() code to generate summary data based on picking columns starting with “c” via the function filter_names(). The pipe operator must be used, and also to subset() function to match the columns.

```
target <- filter_names(names(mtcars),"c")
target
```

```
[1] "cyl" "carb"
```

The resulting vector after the processing should look like this.

```
s_d
```

```
      cyl    carb
6.1875 2.8125
```

Process and visualise the data from eirgrid17 (25 marks)

Load the eirgrid17 tibble from aimsir17.

```
library(aimsir17)
library(ggplot2)
eirgrid17
```

```
# A tibble: 35,040 x 15
```

	year	month	day	hour	minute	date	NIGeneration	NIDemand
	<dbl>	<dbl>	<int>	<int>	<int>	<dtm>	<dbl>	<dbl>
1	2017	1	1	0	0	2017-01-01 00:00:00	889.	776.
2	2017	1	1	0	15	2017-01-01 00:15:00	922.	770.
3	2017	1	1	0	30	2017-01-01 00:30:00	908.	761.
4	2017	1	1	0	45	2017-01-01 00:45:00	919.	743.
5	2017	1	1	1	0	2017-01-01 01:00:00	882.	749.
6	2017	1	1	1	15	2017-01-01 01:15:00	849.	742.
7	2017	1	1	1	30	2017-01-01 01:30:00	843.	726.
8	2017	1	1	1	45	2017-01-01 01:45:00	809.	709.
9	2017	1	1	2	0	2017-01-01 02:00:00	797.	697.
10	2017	1	1	2	15	2017-01-01 02:15:00	755.	684.

```
# i 35,030 more rows
```

```
# i 7 more variables: NIWindAvailability <dbl>, NIWindGeneration <dbl>,  
#   IEGeneration <dbl>, IEDemand <dbl>, IEWindAvailability <dbl>,  
#   IEWindGeneration <dbl>, SNSP <chr>
```

Extract the data for wind energy generated for January 2017, making use of the pipe operator.

```
w_en_jan
```

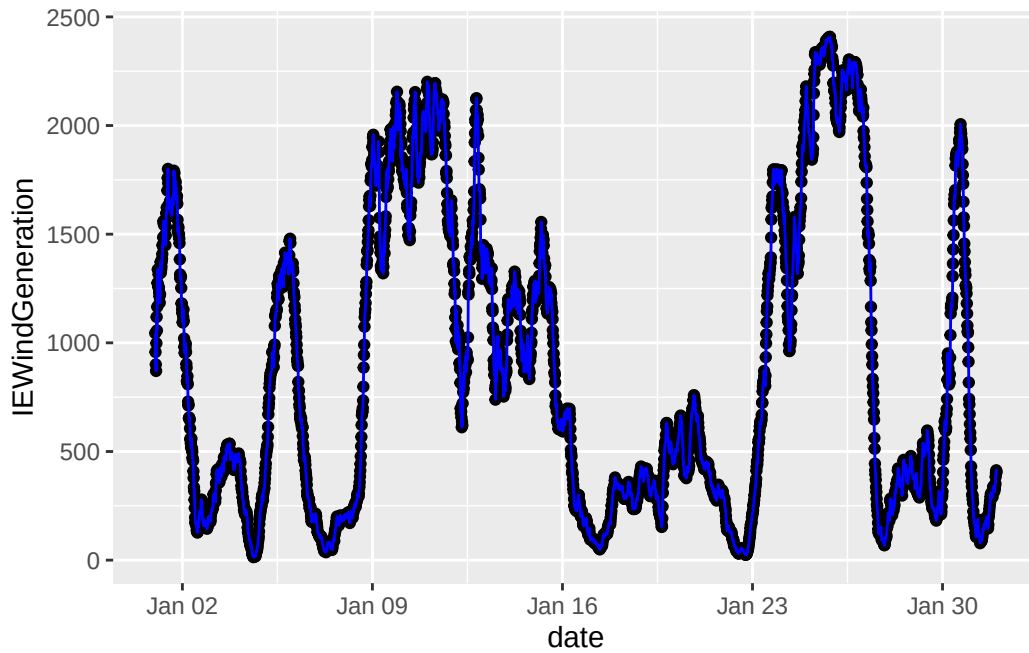
```
# A tibble: 2,976 x 2
```

	date	IEWindGeneration
	<dtm>	<dbl>
1	2017-01-01 00:00:00	1045.
2	2017-01-01 00:15:00	958.
3	2017-01-01 00:30:00	900.
4	2017-01-01 00:45:00	871.
5	2017-01-01 01:00:00	998.
6	2017-01-01 01:15:00	1119.
7	2017-01-01 01:30:00	1057.

```
8 2017-01-01 01:45:00      1166.  
9 2017-01-01 02:00:00      1276.  
10 2017-01-01 02:15:00     1338.  
# i 2,966 more rows
```

Plot the data as a time series.

p1



Process and visualise the data from observations (25 marks)

The aim here is to investigate a potential association between mean sea level pressure and wind speed during winter (months 11, 12, and 1) at the weather station “MACE HEAD”.

First, generate the dataset that contains the data.

```
# Show the first 6 values
head(ob_mar_mh)
```

```
# A tibble: 6 x 4
  date          station    msl  wdsp
<dtm>          <chr>    <dbl> <dbl>
1 2017-01-01 00:00:00 MACE HEAD 1023.    17
2 2017-01-01 01:00:00 MACE HEAD 1023.    17
3 2017-01-01 02:00:00 MACE HEAD 1023.    14
4 2017-01-01 03:00:00 MACE HEAD 1023.    15
5 2017-01-01 04:00:00 MACE HEAD 1024.    12
6 2017-01-01 05:00:00 MACE HEAD 1024    13
```

```
# Show the final 6 values
tail(ob_mar_mh)
```

```
# A tibble: 6 x 4
  date          station    msl  wdsp
<dtm>          <chr>    <dbl> <dbl>
1 2017-12-31 18:00:00 MACE HEAD  988.    27
2 2017-12-31 19:00:00 MACE HEAD  989.    30
3 2017-12-31 20:00:00 MACE HEAD  989.    29
4 2017-12-31 21:00:00 MACE HEAD  989.    28
5 2017-12-31 22:00:00 MACE HEAD  989.    32
6 2017-12-31 23:00:00 MACE HEAD  990.    30
```

From this dataset, find the observations with the minimum and maximum wind speeds (use one call to subset for this).

```
ws_min_max
```

```
# A tibble: 8 x 4
  date          station    msl  wdsp
<dtm>          <chr>    <dbl> <dbl>
```

1	2017-01-02	19:00:00	MACE HEAD	1035.	1
2	2017-01-02	20:00:00	MACE HEAD	1035.	1
3	2017-11-02	19:00:00	MACE HEAD	1016.	1
4	2017-11-02	21:00:00	MACE HEAD	1016	1
5	2017-11-12	23:00:00	MACE HEAD	1030.	1
6	2017-11-14	23:00:00	MACE HEAD	1023.	1
7	2017-11-18	15:00:00	MACE HEAD	1024.	1
8	2017-12-31	02:00:00	MACE HEAD	979.	46

Finally, plot the relationship between msl and wind speed, and also fit to this a linear model (using ggplot2)

p2

```
`geom_smooth()` using formula = 'y ~ x'
```

